Area Between Curves

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area between a curve and the x-axis**

Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.

Roots at $x = \pm 2$. Area $= \int_{-2}^{2} (4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

2 **[Foundational]** • **Target: Area between two curves (vertical strips)**

Find the area between $y = x^2$ and $y = x + 2$.

Intersect: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$: $x = -1, 2$. On this interval $x + 2 \geq x^2$. Area $= \int_{-1}^{2} [(x + 2) - x^2]dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2} = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) = \frac{10}{3} - \left(-\frac{7}{6}\right) = \frac{9}{2}$.

3 **[Foundational]** • **Target: Area between two curves (vertical strips)**

Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} \geq x^2$. Area $= \int_0^1 (\sqrt{x} - x^2)dx = \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

4 **[Foundational]** • **Target: Regions requiring horizontal strips**

Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y.

Intersect: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0$: $y = -1, 2$. On this interval $y + 2 \geq y^2$. Area $= \int_{-1}^{2} [(y + 2) - y^2]dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2} = \left(2 + 4 - \frac{8}{3}\right) - \left(\frac{1}{2} - 2 + \frac{1}{3}\right) = \frac{9}{2}$.

5 **[Foundational]** • **Target: Regions with multiple intersection points**

Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

On $[0, \frac{\pi}{4}]$, $\cos x \geq \sin x$; on $[\frac{\pi}{4}, \pi]$, $\sin x \geq \cos x$. Total area $= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi} (\sin x - \cos x)\,dx$. First: $[\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1$. Second: $[-\cos x - \sin x]_{\pi/4}^{\pi} = (1 - 0) - (-\sqrt{2}) = 1 + \sqrt{2}$. Total $= (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2} \approx 2.83$.

6 **[Foundational]** • **Target: Area using the more efficient variable of integration**

Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.

Solving for x: $x = 4 - y^2$, bounded by $x = 0$ (i.e. $y = \pm 2$). Integrating with respect to y is easier: Area $= \int_{-2}^{2} (4 - y^2)\,dy = \left[4y - \frac{y^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

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7 **[Foundational]** • **Target: Setting up but not evaluating (AP-style)**

Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.

On $[1, 2]$, $e^x \geq \ln x$ (since e^x grows much faster). Area $= \int_1^2 (e^x - \ln x)dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Setting up but not evaluating (AP-style)**

Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x-axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

On $[-2, 0]$, $x^3 - 4x \geq 0$; on $[0, 2]$, $x^3 - 4x \leq 0$. Total area

$= \int_{-2}^{0} (x^3 - 4x)\,dx + \int_0^2 -(x^3 - 4x)\,dx = \int_{-2}^{0} (x^3 - 4x)\,dx - \int_0^2 (x^3 - 4x)\,dx.$

9 **[Intermediate]** • **Target: Area between curves and a table of values**

Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)]\,dx$.

Let $h(x) = f(x) - g(x)$: $h(0) = 7$, $h(2) = 4$, $h(4) = 4$, $h(6) = 6$. Trapezoidal Rule with width 2:

$T = 2\left[\frac{7+4}{2} + \frac{4+4}{2} + \frac{4+6}{2}\right] = 2[5.5 + 4 + 5] = 2(14.5) = 29.$

10 **[Intermediate]** • **Target: Visualizing area between two curves**

The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.

The line $y = x + 2$ lies above the parabola $y = x^2$ throughout $(-1, 2)$, matching the earlier setup $\int_{-1}^{2} [(x + 2) - x^2]dx = \frac{9}{2}$.

11 **[Intermediate]** • **Target: A capstone area problem**

Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^{0} (x^3 - x)\,dx + \left|\int_0^1 (x^3 - x)\,dx\right|$.
$\int_{-1}^{0} (x^3 - x)\,dx = \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} = 0 - \left(\frac{1}{4} - \frac{1}{2}\right) = \frac{1}{4}$. By symmetry the second piece is also $\frac{1}{4}$. Total area $= \frac{1}{2}$.

12 **[Intermediate]** • **Target: Area between an exponential curve and a line**

Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x, tangent only at $x = 0$).

Area $= \int_0^2 [e^x - (x + 1)]dx = \left[e^x - \frac{x^2}{2} - x\right]_0^2 = (e^2 - 2 - 2) - (1 - 0 - 0) = e^2 - 5.$

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13 **[Intermediate]** • **Target: An applied area-between-curves problem**

Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

Intersection: $50 - 2t = 30 + t \Rightarrow 20 = 3t \Rightarrow t = \frac{20}{3} \approx 6.67$. For $t < \frac{20}{3}$, $P_1 > P_2$; for $t > \frac{20}{3}$, $P_2 > P_1$. Total absolute difference $= \left| \int_0^{20/3} (P_1 - P_2)\,dt \right| + \left| \int_{20/3}^{10} (P_2 - P_1)\,dt \right|$, which can be computed piecewise using $P_1 - P_2 = 20 - 3t$.

14 **[Intermediate]** • **Target: Regions requiring horizontal strips**

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

Find intersections with $y = 0$: $3x - x^2 = 0 \Rightarrow x(3 - x) = 0$, so $a = 0$ and $b = 3$. Revolving about y-axis: shell radius $r(x) = x$, shell height $h(x) = 3x - x^2$. $V = 2\pi \int_0^3 x(3x - x^2)\,dx = 2\pi \int_0^3 (3x^2 - x^3)\,dx = 2\pi \left[x^3 - \frac{x^4}{4} \right]_0^3$ $= 2\pi \left(27 - \frac{81}{4} \right) = 2\pi \left(\frac{27}{4} \right) = \frac{27\pi}{2}$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Regions with multiple intersection points** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

Find intersection points: $x^2 = 2x - x^2 \Rightarrow 2x^2 - 2x = 0 \Rightarrow 2x(x - 1) = 0$, so $x = 0$ and $x = 1$. On $[0, 1]$, test $x = 0.5$: $2(0.5) - (0.5)^2 = 0.75 > (0.5)^2 = 0.25$, so $y = 2x - x^2$ is the upper curve.

$A = \int_0^1 [(2x - x^2) - x^2]\,dx = \int_0^1 (2x - 2x^2)\,dx = \left[x^2 - \frac{2x^3}{3} \right]_0^1 = 1 - \frac{2}{3} = \frac{1}{3}$

16 **[Challenging]** • **Target: Area using the more efficient variable of integration** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

1. Express x as a function of y: $y = \ln x \Rightarrow x = e^y$. 2. Apply the volume of revolution about the y-axis formula: $V = \pi \int_c^d [x(y)]^2\,dy = \pi \int_0^1 (e^y)^2\,dy = \pi \int_0^1 e^{2y}\,dy$. 3. Integrate: $V = \pi \left[\frac{e^{2y}}{2} \right]_0^1 = \frac{\pi}{2}(e^2 - e^0) = \frac{\pi(e^2 - 1)}{2}$.

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17 [Challenging] • Target: Setting up but not evaluating (AP-style) • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

1. Define the difference variable $W = X - Y$. The condition $X > Y$ is equivalent to $W > 0$. 2. Since X and Y are independent normals, W is normally distributed: Mean: $\mu_W = E[X] - E[Y] = 50 - 45 = 5$. Variance: $\sigma_W^2 = \text{Var}(X) + \text{Var}(Y) = 16 + 9 = 25 \Rightarrow \sigma_W = 5$. Thus $W \sim N(5, 25)$. 3. Compute $P(W > 0)$: $z = \frac{0-5}{5} = -1.0$. $P(W > 0) = P(Z > -1.0) = P(Z < 1.0) = \Phi(1.0) \approx 0.8413 \approx 0.841$.

18 [Challenging] • Target: Area between curves and a table of values • Cross-Lesson: 2.01 The Definition of the Derivative

Level 1 · FoundationsEvaluate the improper integral $\int_1^\infty \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

Rewrite as a limit: $\lim_{b\to\infty} \int_1^b x^{-3}\,dx = \lim_{b\to\infty} \left[-\frac{1}{2x^2}\right]_1^b = \lim_{b\to\infty} \left(-\frac{1}{2b^2} - \left(-\frac{1}{2(1)^2}\right)\right)$. Since $\lim_{b\to\infty} \frac{1}{2b^2} = 0$: $0 - \left(-\frac{1}{2}\right) = \mathbf{\frac{1}{2}}$. The integral converges to $\frac{1}{2}$.

19 [Challenging] • Target: Visualizing area between two curves • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Level 1 · FoundationsEvaluate the indefinite integral $\int \arctan(x)\,dx$.

Use LIATE: $u = \arctan(x) \Rightarrow du = \frac{1}{1+x^2}\,dx$, and $dv = dx \Rightarrow v = x$. Applying parts: $\int \arctan(x)\,dx = x\arctan(x) - \int \frac{x}{1+x^2}\,dx$. For the second integral, let $w = 1 + x^2 \Rightarrow dw = 2x\,dx$: $\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\ln(1+x^2)$. Thus: $\mathbf{x\arctan(x) - \frac{1}{2}\ln(1+x^2) + C}$.

20 [Challenging] • Target: A capstone area problem • Cross-Lesson: 3.01 The Chain Rule

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

Apply the volume of revolution formula: $V = \pi\int_0^4 y^2\,dx = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = \pi\left(\frac{16}{2} - 0\right) = 8\pi$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Area between a curve and the x-axis

Find the total area between $y = x^3 - 36x$ and the x-axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.

a. 324

b. 654

c. 108

d. 648 ✓

By symmetry, total area $= 2\int_0^6 (6^2x - x^3)dx = 2\left[\frac{6^2x^2}{2} - \frac{x^4}{4}\right]_0^6 = 2\left(\frac{1296}{2} - \frac{1296}{4}\right) = 2\left(\frac{1296}{4}\right) = \frac{1296}{2} = 648$.

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22 [Foundational] • Target: Area between two curves (vertical strips)

Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

a. 39

b. 29

c. 31

d. 30 ✓

$T = 1\left[\frac{8+15}{2} + \frac{15+6}{2} + \frac{6+10}{2}\right] = 1\left[\frac{8+10}{2} + 15 + 6\right] = 1(9 + 15 + 6) = 30.$

23 [Foundational] • Target: Regions requiring horizontal strips

Find the area between $y = e^{-2x}$ and $y = -2x + 1$ on $[0, 1]$ (note $e^{-2x} \geq -2x + 1$ for all x, tangent only at $x = 0$), rounded to two decimal places.

a. 0.43 ✓

b. 0.52

c. 0.35

d. 1.43

Area $= \int_0^1 [e^{-2x} - (-2x + 1)]dx = \left[\frac{e^{-2x}}{-2} - \frac{-2x^2}{2} - x\right]_0^1 \approx 0.43.$

24 [Foundational] • Target: Regions with multiple intersection points

Consider the region bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 0$ and $x = b$, where $b > 0$. If the area of this region is $\frac{\pi}{3}$, what is the value of b?

a. $\sqrt{2}$

b. $\frac{1}{\sqrt{3}}$

c. $\sqrt{3}$ ✓

d. 2

The area of the region is given by the definite integral of the function from $x = 0$ to $x = b$. The integral of $\frac{1}{x^2+1}$ is $\arctan(x)$. Therefore, the area is given by: $\text{Area} = \int_0^b \frac{1}{x^2+1}dx = [\arctan(x)]_0^b = \arctan(b) - \arctan(0) = \arctan(b)$ We are given that the area is $\frac{\pi}{3}$. So, we have: $\arctan(b) = \frac{\pi}{3}$ To find b, we take the tangent of both sides: $b = \tan\left(\frac{\pi}{3}\right)$ We know that.

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25 [Foundational] • Target: Area using the more efficient variable of integration

Consider the region bounded by the curve $y = x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. What is the area of this region?

a. $\frac{26}{3}$ ✓

b. $\frac{28}{3}$

c. $\frac{25}{3}$

d. $\frac{27}{3}$

The area of a region bounded by a curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$ is given by the definite integral $\int_a^b f(x)\,dx$. In this case, $f(x) = x^2$, $a = 1$, and $b = 3$. The area A is given by: $A = \int_1^3 x^2\,dx$ To evaluate this integral, we find the antiderivative of x^2, which is $\frac{x^3}{3}$. Now, we apply the Fundamental Theorem of Calculus.

26 [Foundational] • Target: Setting up but not evaluating (AP-style)

Consider the region R in the first quadrant bounded by the curve $y = e^{-x^2}$, the x-axis, and the line $x = a$, where $a > 0$. A solid is generated by revolving this region R about the y-axis. If the volume of this solid is $\frac{\pi}{2}$, determine the value of a.

a. $\ln(2)$

b. $\sqrt{\ln(2)}$ ✓

c. $e^{\sqrt{2}}$

d. $\sqrt{2}$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and the lines $x = a$ and $x = b$ about the y-axis can be found using the shell method. The formula for the volume V is given by $V = \int_a^b 2\pi x f(x) dx$. In this problem, $f(x) = e^{-x^2}$, and the region is bounded by $y = e^{-x^2}$, the x-axis, and the line $x = a$ in the first quadrant. Thus, the limits of integration are from $x = 0$ to $x = a$. The volume of the solid is given by.

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27 [Foundational] • Target: Area between curves and a table of values

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. **8/3** ✓

d. 16/3

The region R is bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$. We are revolving this region about the vertical line $x = 5$. We will use the method of cylindrical shells. The radius of a cylindrical shell at a given x is the distance from the axis of revolution $x = 5$ to x, which is $5 - x$. The height of the shell is the difference in the y-values of the bounding curves, which is $\sqrt{x} - 0 = \sqrt{x}$. The thickness of the shell is dx. The volume of a single cylindrical shell is.

28 [Foundational] • Target: Visualizing area between two curves

$\int_a^b [f(x) - g(x)]dx$ gives area between curves when:

a. $f > g$ and $f < g$ alternates

b. $f > 0$ and $g < 0$

c. $f(x) \geq g(x)$ **on** $[a, b]$ ✓

d. always

Applying Applications of Integration principles, Area when $f \geq g$. Therefore, the verified correct option is $f(x) \geq g(x)$ on $[a, b]$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A capstone area problem

Find the volume when the area between $y = \sec x$ and $y = \cos x$ (from $-\pi/4$ to $\pi/4$) is rotated about the x-axis:

a. $\pi^2/4$

b. $2\pi(\pi/4 - 1/2)$

c. $\pi(2 - \pi/2)$ ✓

d. $\pi(\sqrt{2} - 1)$

Applying Applications of Integration principles, Washer method. Therefore, the verified correct option is $\pi(2 - \pi/2)$.

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30 [Intermediate] • Target: Area between an exponential curve and a line

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$ ✓

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

$A = \frac{1}{2}\int (r_2^2 - r_1^2)d\theta$.

31 [Intermediate] • Target: An applied area-between-curves problem

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)|\,dx$ ✓

b. $\int_a^b f(x)\,dx$ only when $f \ge 0$

c. $f(b) - f(a)$

d. $\int_a^b f(x)^2\,dx$

Always use absolute value for area.

32 [Intermediate] • Target: Area between a curve and the x-axis

The region bounded by the curve $y = x^2 - 4$ and the x-axis is rotated about the x-axis. If the volume of the resulting solid is $V = \pi\int_a^b (x^2 - 4)^2 dx$, find the value of $b - a$.

a. 2

b. **4** ✓

c. 8

d. 16

The region is bounded by the curve $y = x^2 - 4$ and the x-axis. To find the limits of integration, we need to find the x-intercepts of the curve, where $y = 0$. Setting $x^2 - 4 = 0$, we get $x^2 = 4$, which gives $x = \pm 2$. Therefore, the bounds of integration are $a = -2$ and $b = 2$. The volume of the solid of revolution is given by $V = \pi\int_a^b [f(x)]^2 dx$. In this case, $f(x) = x^2 - 4$, and the limits are $a = -2$ and $b = 2$. The question asks for the value of $b - a$. $b - a = 2 - (-2) = 2 + 2 = 4$.

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33 [Intermediate] • Target: Area between two curves (vertical strips)

Area below x-axis from -2 to 0 of $f(x) = x^2 - 4$:

a. $-\frac{16}{3}$

b. $\frac{8}{3}$

c. $\frac{16}{3}$ ✓

d. 4

Since $x^2 - 4 \leq 0$ on $[-2, 0]$, the area is $\int_{-2}^{0}(4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{0} = 0 - \left(-8 + \frac{8}{3}\right) = \frac{16}{3}$.

34 [Intermediate] • Target: Regions requiring horizontal strips

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ **and** $dx/dte0$ ✓

c. $t = 0$

d. $dy/dt = dx/dt$

Applying Parametric and Polar principles, Horizontal tangent: $dy/dt = 0$. Therefore, the verified correct option is $dy/dt = 0$ and $dx/dte0$.

35 [Intermediate] • Target: Regions with multiple intersection points

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ **and** $dy/dte0$ ✓

c. $dx/dt = dy/dt$

d. $t = 0$

Applying Parametric and Polar principles, Vertical tangent: $dx/dt = 0$. Therefore, the verified correct option is $dx/dt = 0$ and $dy/dte0$.

36 [Intermediate] • Target: Area using the more efficient variable of integration

Area between $y = x$ and $y = x^2$: intersect at $x =$

a. 0 **and** 1 ✓

b. 1 and 2

c. 0 and 2

d. -1 and 1

Applying Applications of Integration principles, $x = x^2 \Rightarrow x(x - 1) = 0$. Therefore, the verified correct option is 0 and 1.

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37 [Intermediate] • Target: Setting up but not evaluating (AP-style)

Signed area and winding number: the winding number of curve C around origin is $\frac{1}{2\pi}\oint_C \frac{x\,dy - y\,dx}{x^2+y^2} =$

a. always 1

b. **an integer ✓**

c. always 0

d. always π

Applying Applications of Integration principles, Winding number is an integer. Therefore, the verified correct option is an integer.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Area between curves and a table of values • Cross-Lesson: 2.01 The Definition of the Derivative

Volume of the solid of intersection of two cylinders $x^2 + z^2 \le 1$ and $y^2 + z^2 \le 1$:

a. $\frac{8}{3}$

b. $\frac{16}{3}$ ✓

c. π

d. 4

Applying Applications of Integration principles, Steinmetz solid: $16/3$. Therefore, the verified correct option is $\frac{16}{3}$.

39 [Challenging] • Target: Visualizing area between two curves • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$ ✓

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

The curves meet at $x = 0$ and $x = 1$. Shells: $V = 2\pi\int_0^1 x(-x^3 + x)\,dx = \frac{4\pi}{15}$.

40 [Challenging] • Target: A capstone area problem • Cross-Lesson: 3.01 The Chain Rule

Area of triangle with vertices at origin, $(3, 0)$, $(0, 4)$ using integration:

a. 3

b. 12

c. 6 ✓

d. π

Applying Applications of Integration principles, $A = \frac{1}{2}(3)(4) = 6$. Therefore, the verified correct option is 6.

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41 **[Challenging]** • **Target: Area between an exponential curve and a line** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4 ✓

d. 8

Applying Fundamental Theorem principles, $2\int_0^2 x\,dx = 2 \cdot 2 = 4$. Therefore, the verified correct option is 4 .

42 **[Challenging]** • **Target: An applied area-between-curves problem** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Moments of inertia: second moment of area $I_x = \int y^2\,dA$ for rectangle $b \times h$ about its centroidal axis:

a. $bh^3/12$ ✓

b. $bh^3/3$

c. $bh^2/12$

d. $b^2h/12$

Applying Applications of Integration principles, $I_x = bh^3/12$. Therefore, the verified correct option is $bh^3/12$.

43 **[Challenging]** • **Target: Area between a curve and the x-axis** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Surface area of $y = \sqrt{x}$ revolved about the x-axis from 0 to 4:

a. $\frac{\pi}{3}(17\sqrt{17} - 1)$

b. $\pi(\sqrt{17} - 1)$

c. 8π

d. $\frac{\pi}{6}(17\sqrt{17} - 1)$ ✓

$S = 2\pi\int_0^4 \sqrt{x}\sqrt{1 + 1/(4x)}\,dx = \pi\int_0^4 \sqrt{4x+1}\,d(x) \to \frac{\pi}{6}(17\sqrt{17} - 1)$.

44 **[Challenging]** • **Target: Area between two curves (vertical strips)** • **Cross-Lesson: 5.01 Extrema and the Mean Value Theorem**

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$ ✓

d. $\pi e^{4\pi}$

$A = \frac{1}{2}\int_0^{2\pi} e^{2\theta} d\theta = \frac{1}{4}[e^{2\theta}]_0^{2\pi} = \frac{e^{4\pi}-1}{4}$.

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45 [Challenging] • Target: Regions requiring horizontal strips • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Surface area of Gabriel's horn $y = 1/x$, $x \geq 1$, rotated about x-axis:

a. **diverges ✓**

b. π

c. 2π

d. finite

$S = 2\pi\int_1^\infty (1/x)\sqrt{1 + 1/x^4}\,dx > 2\pi\int_1^\infty (1/x)dx$: diverges (Painter's paradox).

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